

AI 3000: Reinforcement Learning

Tutorial 0

1 Simulating a Markov chain

We would like to simulate a discrete-time Markov chain over a finite set $\mathcal{X}$ with initial state X_0 distributed according to a pmf μ ,¹ and with a time-homogeneous transition probability matrix P , i.e.

$$X_0 \sim \mu, \quad \mathbb{P}(X_{n+1} = y \mid X_n = x) = P(x, y).$$

1.1 We need to produce samples from μ to initialize the chain. Assuming we can produce samples from $\text{Unif}(0, 1)$, how can we generate $X_0 \sim \mu$?

Does your method generalize to cases where $\mathcal{X}$ is uncountable, and we have a density p^* ?

1.2 (Rejection Sampling) Rejection sampling assumes that we have a target density p^* over $\mathbb{R}^d$ which we can evaluate and would like to sample from, and some base density p_0 that we can both evaluate and sample from. This base density is such that there exists some positive real number M so that for all $x \in \mathbb{R}^d$

$$p^*(x) \leq M p_0(x).$$

The rejection sampling procedure is then as follows:

1. Sample a candidate $X \sim p_0$.
2. Sample $U \sim \text{Unif}(0, 1)$ independently of X .
3. Accept the candidate if $U \leq \frac{p^*(X)}{M p_0(X)}$.

Show that an accepted candidate X has density p^* , i.e. show that the conditional density $f_{X|A=1}$ is p^* .

1.3 Now that we are equipped to sample from arbitrary pmfs, can you simulate the Markov chain described by (μ, P) ?³

2 Markov chain proofs

2.1 Let $\{X_n\}_{n \geq 0}$ be a discrete-time Markov chain over a finite state space $\mathcal{X}$, and $n \in \mathbb{N}$ be arbitrary. At time n , let us denote the history of the Markov chain to be $H_n = (X_{n-1}, \dots, X_0)$.⁴ Then we defined the Markov property to be

$$\mathbb{P}(X_{n+1} = y \mid X_n = x, H_n = h_n) = \mathbb{P}(X_{n+1} = y \mid X_n = x)$$

where $y \in \mathcal{X}$, $h_n = (x_{n-1}, \dots, x_0)$, and $(x_0, \dots, x_{n-1}, x_n)$ is a trajectory that the Markov chain can take with positive probability.⁵ Prove that this condition implies that

$$\mathbb{P}(X_m = y \mid X_n = x, H_n = h_n) = \mathbb{P}(X_m = y \mid X_n = x)$$

for every $m > n$.

1. In class, we assumed the initial state to be deterministic. Do we lose anything by assuming it to be stochastic instead?

2. Given an arbitrary p^* , can you find such a base density p_0 ? What if p^* has bounded support?

3. Hint: Every row of P is a pmf.

4. This is purely for convenient notation.

5. Why is this necessary to state?

2.2 (Random walks) Let $Z_1, Z_2, \dots \stackrel{\text{iid}}{\sim} \text{Rademacher}(p)$.⁶ Then we proved in class that the process $\{S_n\}_{n \geq 0}$ is a Markov chain where

$$S_n := \sum_{i=1}^n Z_i.$$

(The empty sum is 0 by convention so that $S_0 = 0$)

Explicitly calculate the marginal pmf of S_n , i.e. $\mathbb{P}(S_n = x)$.

2.3 We define two processes $\{A_n\}_{n \geq 0}$ and $\{M_n\}_{n \geq 0}$ derived from S_n .

$$A_n := |S_n|, \quad M_n := \max\{0, S_n\}.$$

Are A and M also Markov chains? If so prove it, and if not given an example where the Markov property fails.

⁶ $X \sim \text{Rademacher}(p)$ means that

$$\mathbb{P}(X = 1) = p = 1 - \mathbb{P}(X = -1).$$

Can you relate the Rademacher and Bernoulli distributions? What do you know about the sum of Bernoulli random variables?

3 Markov Decision Processes

We would like to model the following problems as Markov Decision Processes. Write down the state space $\mathcal{X}$, action space $\mathcal{A}$, reward alphabet $\mathcal{R}$.⁷ Are any of these problems trivial, i.e. are there any obvious solutions?

⁷ There may not be an unambiguous answer to any of these; feel free to explore different possibilities.

3.1 Consider the single-player Snakes and Ladders game. With only a single die, we are completely at the mercy of Nature. However, assume we have K dice each with a different pmf. Now we can influence the outcomes by choosing which die to pick. We would now like to finish the game (by reaching 100) as soon as possible.

3.2 A farmer is renting a fish farm with some maximum capacity and a fixed annual rent. Every year the farmer must decide how many fish to harvest, and leave the rest to spawn for next year's harvest. The population growth is stochastic and not known to the farmer, but clearly the more fish are left to spawn, the better the chance of a larger population next year, but the more the farmer has to pay for feed. On the other hand, the more fish are harvested, the more profit the farmer makes this year. The farmer would like to maximize their profit over the next several years.

3.3 An astronomer is trying to find supernovae. It is theoretically expected that supernovae are more likely to occur in galaxies with high star-formation rates, and so the astronomer has compiled a catalogue of such galaxies to monitor for supernovae. However due to limited telescope time, the astronomer can only observe one galaxy every night. She would like to find as many supernovae as possible so that further studies can be done on them using larger ground-based or space-based telescopes.